3 12-Lead ECG

HME922 Procedural Station · 10 minutes · CR4 ECG guideline

guideline wording — say it aloud DOPS On the examiner's DOPS marking form ☐ tick when done

KEY MESSAGE

10 electrodes → 12 leads. Find the 4th ICS, place V1–V6, record with patient still, **label the trace immediately**, then read it in 's fixed order.

UNIVERSAL OPENING

- ☐ Hand hygiene — Moment 1
- ☐ Introduce yourself: name + "medical student"
- ☐ Confirm ID — **3 identifiers**
- ☐ Explain: bare chest; some leads go **under the left breast**
- ☐ Ask about **skin/adhesive allergies**; brief history — why the ECG?
- ☐ Allow time for questions
- ☐ Gain **informed verbal consent**; offer a chaperone
- ☐ Private, **warm** room; supine; not over-exposed

MEMORY AID — 5 MOMENTS

5 Moments of Hand Hygiene: 1 before touching patient · 2 before procedure · 3 after body-fluid risk · 4 after touching patient · 5 after touching surroundings

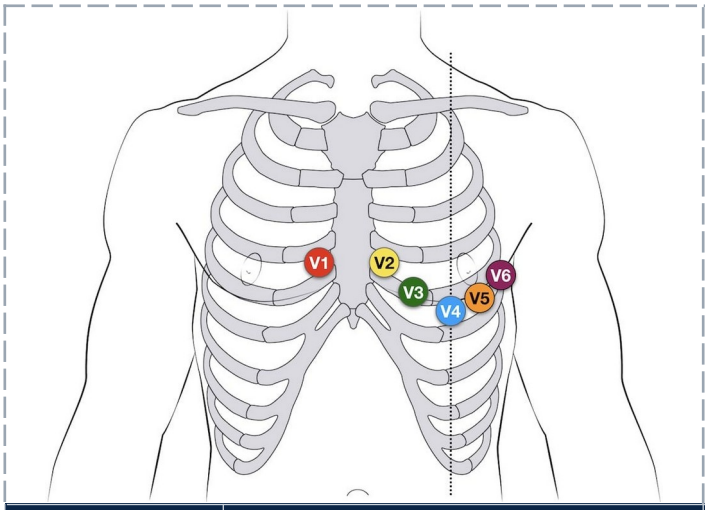
CLINICAL KNOWLEDGE — SAY IT

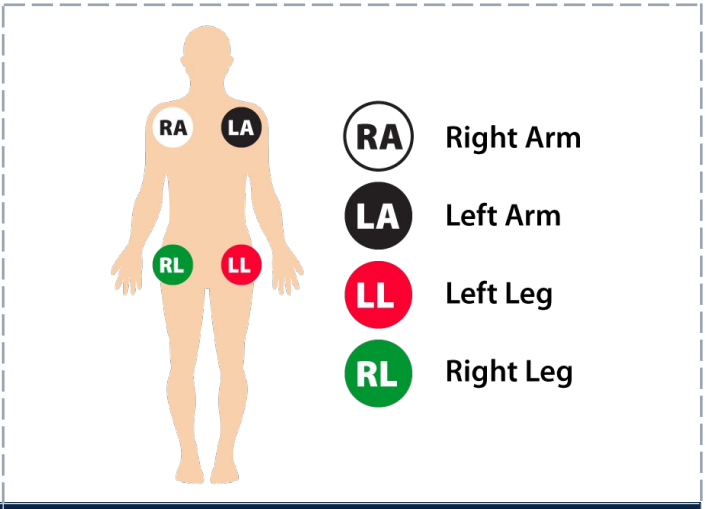
- ☐ Indications: chest pain · palpitations · heart failure · hypertension · SOB · congenital heart disease · electrolytes · drugs/toxins · routine medicals
- ☐ Detects: arrhythmia, ischaemia/AMI, conduction block (BBB), K+ disturbance, PE
- ☐ No alcohol, tobacco, caffeine or vigorous exercise in previous **30 minutes**
- ☐ RL = earth. I, II, III = bipolar (Einthoven). aVR, aVL, aVF = augmented. V1–V6 = unipolar, horizontal plane. 3/5-lead = rhythm monitoring only

PREPARE

- ☐ ECG machine charged + paper; paper towel; disposable clippers + sharps bin; sandpaper; electrodes; 10 labelled leads; tape
- ☐ Wipe sweat; **clip** hair (don't shave); sandpaper if needed

ELECTRODE PLACEMENT





Electrode	Position	Remember
V1	4th ICS, **RIGHT** sternal border	V1 & V2 = 4th ICS, either side

V2	4th ICS, **LEFT** sternal border	of the sternum
V4	5th ICS, mid-clavicular line	**Place V4 before V3**
V3	Midway V2 ↔ V4	the bridge
V5	Anterior axillary line, level of V4	5, 6 slide out sideways
V6	Mid-axillary line, level of V4	in a straight line
RA / LA	Inside right / left wrist	
RL / LL	Inside right / left ankle	RL = earth

MEMORY AID — PLACING CHEST LEADS

Find the **angle of Louis** → 2nd rib beside it → slide down to the **4th ICS** → V1 right, V2 left.

"**4-4-5**": V1 & V2 in the **4**th, V4 in the **5**th; V3 between; V5 & V6 follow V4 out to the side.

Limb colours (AHA, as in diagram): "**White on right, smoke (black) over fire (red), clouds over grass**". IEC machines: RA red, LA yellow, LL green, RL black — "**R**ide **Y**our **G**reen **B**ike" clockwise. Always check the lead labels.

Posterior (V7–V9) and right-sided leads are **not examinable**.

RECORD

1. Patient relaxed, still, not talking → **RECORD**
2. Rhythm strip prints (usually lead II)
3. Switch off, remove electrodes, patient dresses; wipe leads; machine back on charge
4. **Label the trace now:** name · DOB · date + **time** · chest pain / pain-free · vital signs · your name + signature

INTERPRET — ORDER

MEMORY AID — "Remember: Rate Rhythm, P, PR, QRS, ST, T, QT — then Compare"

Say the **patient details + presentation** first, finish by **comparing with an old ECG**.

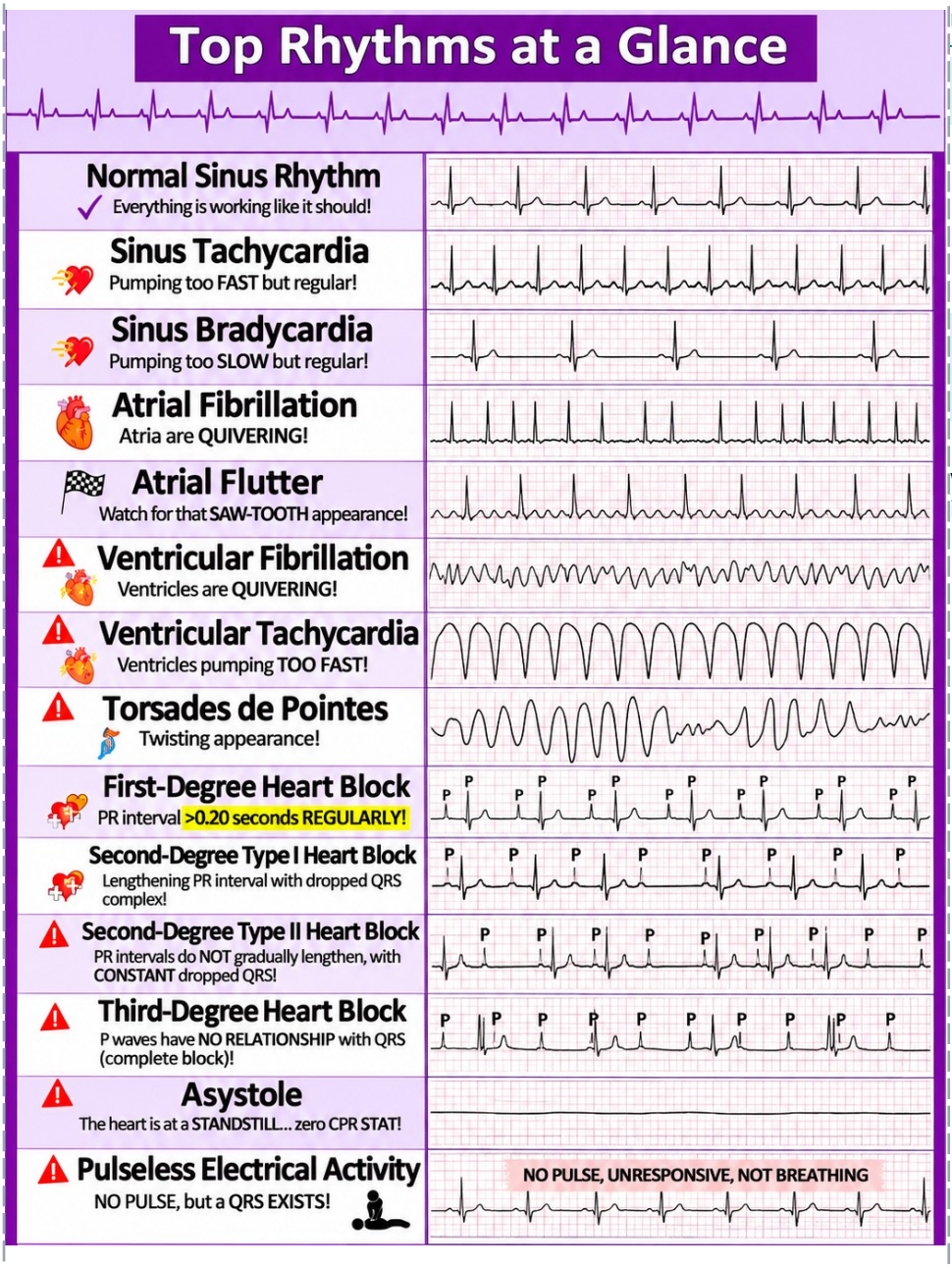
Step	Normal	Look for
Rate	60–100	300 ÷ large squares R–R; or QRS in 10 s × 6 (P waves × 6 = atrial rate)
Rhythm	Regular P–P and R–R	Every P followed by a QRS?
P wave	Upright, < 2.5 mm, < 0.12 s, 1:1	Absent (AF), saw-tooth (flutter)
PR	0.12–0.20 s (3–5 small sq)	Long = heart block
QRS	0.08–0.12 s	Wide (BBB); pathological Q > 0.04 s, > 1 mm, > 25% of R; R-wave progression
ST	Isoelectric	↑ > 1 mm limb / > 2 mm chest in 2 contiguous leads; ↓ ≥ 0.5 mm
T wave	Same direction as QRS; < 5 mm limb, < 10 mm chest	Inversion (TWI V1 can be normal); tall peaked = ↑K+
QTc	< 0.425 s	QTc = QT ÷ √RR; long QT → VT risk

TERRITORIES

MEMORY AID — "I SAL" — Inferior · Septal · Anterior · Lateral

Ischaemia = **repolarisation** (ST ↑ / ↓, T inversion). Infarction = **depolarisation** (pathological Q, small/notched QRS). **Time is muscle** — ~30 min ischaemia → infarction.

Territory	Leads	Artery
Inferior	II, III, aVF	RCA
Septal	V1, V2	LAD
Anterior	V3, V4	LAD
Lateral	I, aVL, V5, V6	LCx



UNIVERSAL CLOSING

- ☐ Clippers into sharps bin; electrodes into general waste
- ☐ Patient dresses; comfortable; bed lowest; call bell
- ☐ Explain next step — doctor will review; report chest pain immediately
- ☐ Hand hygiene — Moments 4/5
- ☐ Trace labelled + signed (check again)
- ☐ Can't answer? "I'll pass that on to someone senior"
- ☐ Examiner asks **"How do you think that went?"** → one thing done well · one to improve · one strategy